HOBench: a literature-based benchmark for hierarchical orthology inference, built by AI agents

[Authors to be added by the human maintainer][1]

1 [Affiliation]

Note on how this work was produced. This study was an experiment in testing whether AI agents can run a bioinformatics research project. An AI agent acted as principal investigator (PI) and directed AI subagents through the literature survey, design, data curation, software, analysis, this manuscript and the accompanying repository. Humans wrote the prompts, supplied papers that the agents requested, and restarted the session after usage limits. No human checked the curated data or the code line by line before this draft. The agent setup is described in Methods; the prompts and the complete process record are in the repository.

Abstract

Benchmarks for orthology inference mostly score pairwise orthology or flat orthologous groups. Hierarchical orthologous groups (HOGs) state, for every taxonomic level, which genes descend from one ancestral gene. Existing benchmarks score HOGs only through these flattened views. We built HOBench, a benchmark that scores HOGs against evolutionary histories asserted in publications. HOBench contains five species panels (102 panel slots, 90 unique species, from UniProt reference proteomes) and 32 curated reference families and collections, including 1,976 yeast whole-genome duplication (WGD) gene sets from the Yeast Gene Order Browser and 498 *Xenopus laevis* homeolog pairs. Each reference is a tree with speciation, duplication, transfer and unresolved nodes, in which literature events are placed on species-tree branches, often as intervals. All four metrics derive from one quantity, the most recent level at which two genes share a HOG: per-level group F1, merge-level accuracy, recovery of literature duplications and family-membership F1. We report each next to trivial baselines. In one pooled test over 13 headline items, OrthoFinder 3 scored higher than FastOMA on per-level F1 (+0.054; Holm-adjusted $p = 0.035$); the other endpoints were not significant. Both tools classified most yeast pseudo-orthologs (paralogs left by reciprocal gene loss) as orthologs. On the yeast WGD they failed in opposite ways: OrthoFinder merged ohnolog copies, FastOMA split each copy's orthologs. In single-seed perturbations of the yeast panel, splitting genes into fragments lowered accuracy more than adding foreign proteins, changing the species tree or removing species. The work was done by AI agents as a test of agent-run research (see note above). The sample of families is small and several design choices are ours rather than community standards, so the results should be read as a prototype. Code, references and tool outputs are available for scoring new methods.

Author summary

Genes in different species are related through speciation and gene duplication. Methods for orthology inference reconstruct these relations, and many now output hierarchical groups: for each ancestor in a species tree, the genes that descend from

one ancestral gene. Existing benchmarks do not test whether these hierarchies are correct at each level. We collected gene families whose evolutionary histories are described in the literature, encoded what the papers assert (and only that), and wrote metrics that test the hierarchy level by level. We used the benchmark to compare two widely used methods and found specific failure modes, such as mistaking paralogs for orthologs after whole-genome duplication. The project was carried out by AI agents under human-written prompts, as an experiment in agent-run research. We describe what the agents did, which of their ideas failed, and which limitations remain.

Introduction

Orthologs are genes that diverged by speciation; paralogs diverged by duplication. A hierarchical orthologous group (HOG) at a taxonomic level L is the set of genes that descend from a single gene in the last common ancestor of the species below L [1, 2]. A set of HOGs at all levels is equivalent to a reconciled gene tree in which duplications are placed on species-tree branches. Several current methods output HOGs, including OrthoFinder [3, 4] and FastOMA [5].

The Quest for Orthologs (QfO) benchmark service [6, 7] evaluates predicted ortholog pairs with tests based on species-tree discordance, reference gene trees and function. OrthoBench scores flat orthogroups against 70 curated reference groups at one level, Bilateria [8, 9]. SwissTree provides expert-curated reconciled gene trees [10], but they are scored through ortholog pairs. Simulated data allow exact HOG evaluation [1, 11], but they lack real-data error sources such as annotation errors. A recent review notes that HOG-specific benchmarking is still missing [12]. We found no real-data benchmark that scores HOGs level by level or checks whether specific duplications described in the literature are recovered.

Building such a benchmark is mainly a problem of data collection. Many gene families have published phylogenies, but the information is spread over text, tables, trees and supplements, uses outdated identifiers, and is often only partly resolved. A useful reference must record exactly what each publication asserts and map it to current proteomes. Everything else must stay unscored. It should also cover the phenomena that make orthology inference hard: whole-genome duplication (WGD) with differential gene loss, horizontal gene transfer (HGT), deep paralogy, promiscuous domains, annotation errors and contamination.

We present HOBench, a benchmark built by AI agents that addresses this problem for 32 families and collections in five species panels. We describe the data collection, the reference encoding and the metrics, and we use the benchmark to compare OrthoFinder and FastOMA. We also report the process: the whole project was an experiment in whether AI agents can run a bioinformatics study, and several of its decisions and failures are informative for that reason.

Results

Overview of the benchmark

HOBench has three parts (Fig 1): species panels with proteomes and species trees, reference families curated from publications, and metrics with controls. Table 1 lists the panels. We chose five panels instead of one large species set because the phenomena of interest occur in different clades and because orthology inference on a single 90-species set exceeded our compute budget (see Methods). Panels are yeasts (Y), prokaryotes (K), eukaryotes with prokaryote outgroups (E), vertebrates (V) and

plants (P). Each panel has a binary species tree whose internal nodes are named levels; topologies come from published phylogenomic studies [13–15].

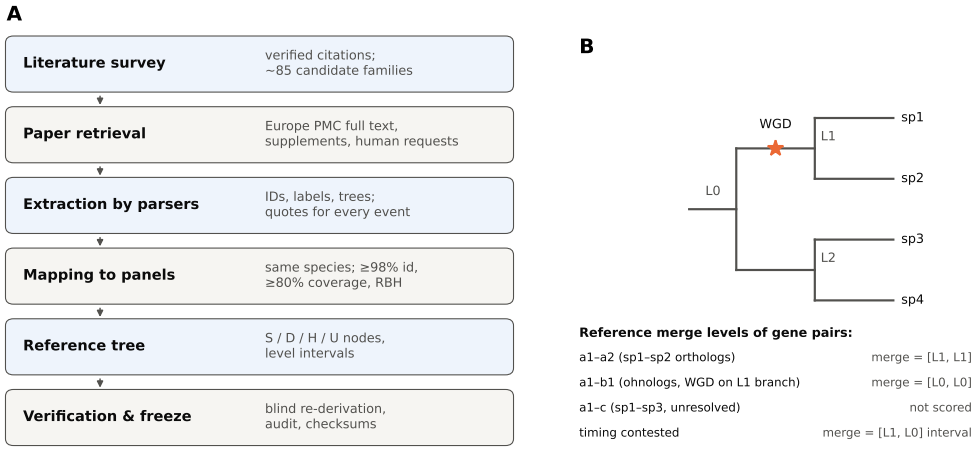


Fig 1. Data collection and the merge-level formalism. (A) Steps from literature to frozen reference families (details in Methods). (B) Toy example. Genes a1 and b1 are ohnologs in species sp1 that arose from a WGD on the branch leading to level L1; a2 is the sp2 ortholog of a1. The merge level of a gene pair in the reference is the most recent level at which the two genes descend from one ancestral gene: L1 for the orthologs a1–a2 and L0 for the ohnologs a1–b1. Pairs that a source does not resolve are not scored. When event timing is contested, the reference allows an interval of merge levels.

Table 1. Species panels. Proteomes are UniProt reference proteomes, release 2026_03 [16], one protein per gene. Subsets are described in Methods.

Panel	Scope	Species	Proteins	Scored on	Main events
Y	Saccharomycetaceae + outgroups	16	85,185	full	allopolyploid WGD, gene loss, HGT
K	Bacteria and archaea	21	68,128	full	ancient paralogs, controls, decoys
E	Eukaryotes + 9 prokaryotes	34	340,217	25,855 subset	pre-LECA paralogs
V	Vertebrates + 2 chordates	17	447,758	70,000 subset	2R, 3R, allotetraploidy
P	Green plants	14	417,784	25,429 subset	plant WGDs, HGT, nested families

Reference families from the literature

Agents screened about 85 candidate families described in the literature and kept those with machine-readable member lists (supplementary tables, labelled trees or curated databases) and at least four panel species after mapping. They excluded sources whose orthology assignments come from tools for orthology inference (OMA, OrthoFinder, Ensembl Compara and names projected from it, PANTHER, TreeFam, eggNOG, OrthoDB, PLAZA, OHNOLOGS), because using them as truth would favour the tools they come from. The final catalogue has 32 entries (Table 2; full list in S1 Table): 13 headline items, 3 control families, 10 non-homologous decoy families, and 6 secondary families that missed the mapping or verification thresholds. Together they contain about 31,800 genes, 85 duplications asserted in the text (tier T1) and 2,136 duplications derived from curated classifications (tier T2).

Table 2. Headline reference families and controls. Members = genes mapped to the panel proteomes. Collections are scored as one item each.

ID	Family	Source	Members	Species	Phenomena
Y01	YGOB pillars (1,976 units)	[17–19]	26,166	13	WGD, loss, hidden paralogy
Y03	URA1 dihydroorotate dehydrogenase	[20, 21]	16	14	HGT from bacteria
E01	Kinesins	[22]	399	19	pre-LECA paralogs, motor domain
E02	ARF/Arl/Sar GTPases	[23]	325	22	pre-LECA paralogs, short domain
E03	Rab GTPases	[24]	388	15	pre-LECA paralogs
E04	Tubulins	[25]	170	24	pre-LECA paralogs
E07c	Adaptor complex scaffolds	[26]	35	14	pre-LECA paralogs
V02	Visual opsins	[27]	41	8	2R, 3R
V03	Hox clusters	[28, 29]	237	5	tandem, 2R, 3R
V07	*X. laevis* L/S homeologs (498 units)	[29]	1,494	2	allotetraploidy
P02	bHLH transcription factors	[30]	435	5	short domain, plant WGDs
P03	GID1 within carboxylesterases	[31]	94	9	nested family
P04	Major intrinsic proteins	[32]	119	4	HGT (NIPs), plant WGDs
K01	Ribosomal proteins (40 units; control)	[33, 34]	651	19	single copy
K02	EF-Tu/EF-G (control)	[35, 36]	47	19	pre-LUCA paralogs
E05	MCM2–9 (control)	[37]	113	18	pre-LECA paralogs
K04	Analogous enzymes (5 pairs; decoys)	[38]	108	5–14	must never merge

Conceptual approach. A reference family records only what its sources assert. Each family is a rooted tree whose leaves are UniProt accessions and whose internal nodes are speciation (S), duplication (D), HGT acquisition (H) or unresolved (U). Duplications are placed on a species-tree branch, or on an interval of branches when the timing is uncertain or contested. Pairs whose lowest common node is U are not scored. We designed the reference this way to avoid the main weakness of benchmarks built from published sources: absence of a statement is not evidence. It also allows partial references, which most families are.

Several histories needed explicit intervals. The yeast WGD probably arose by hybridization of two lineages [19], so ohnolog copies may trace back to one gene in the ancestor of the post-WGD and ZT clades or as deep as the Saccharomycetaceae root; we encoded that range. Using YGOB synteny tracks [17], we identified single-copy genes in two post-WGD species that are paralogs after reciprocal loss (pseudo-orthologs [39]); we scored these as paralogs. *X. laevis* L and S homeologs [29] are a duplication on the *X. laevis* branch. HGT was encoded so that the transferred copy and native copies are never in one HOG within the recipient lineage.

Technical approach. Agents retrieved open-access full texts and supplements from Europe PMC, extracted identifiers and labels with committed per-family parser scripts (no identifiers were typed by hand), mapped them to the panel proteomes and compiled the reference tree. Mapping used UniProt identifier mapping or same-species sequence matches (≥98% identity, ≥80% coverage of both sequences, reciprocal best hit). Every label, event level and tag carries a source locator and a verbatim quote, which a script checks against the stored source text. A different agent re-derived up to 30 sampled assertions per family without seeing the claimed values; 22 of 23 verification groups passed (at most 1 discrepancy) and one (globins) failed and was moved to the secondary set (S2 Text). Families in which fewer than 80% of members mapped uniquely were also kept out of the headline set. We froze the references by checksum before scoring the larger panels. Later corrections are logged with values

before and after the change.

Metrics

For two genes a and b, the *merge level* $m(a, b)$ is the most recent level at which they belong to the same HOG. Orthologs merge at the last common ancestor of their species; paralogs merge above the duplication. The reference gives an interval $[m_{lo}, m_{hi}]$ for each asserted pair; a predicted merge level is correct if it lies inside. The merge-level matrix of a reconciled tree is a symbolic ultrametric [40, 41]. Hence per-level HOG partitions, pairwise relations and duplication timing can all be read from the same matrix. We defined four primary endpoints before scoring the larger panels:

1. **Per-level B-cubed F1** [42]: agreement between predicted and reference HOGs at each level, computed per gene and averaged over levels.

2. **Merge-level accuracy**: for each asserted reference node, the fraction of its gene pairs with a correct predicted merge level, averaged over nodes so that large subfamilies do not dominate.

3. **Event recovery**: a literature duplication is recovered if its copies are in different HOGs at the event level and at least half of each copy's reference orthologs share a HOG. Because both conditions must hold, trivial predictions cannot score.

4. **Family-membership F1**: precision and recall for which genes belong to the family, computed only where errors are defined: in species for which a source states a genome-wide search, against declared decoys, and for separations that the reference asserts.

Secondary measures include the OrthoBench family measures [9], signed level errors (too old or too young), contamination by decoys, and a skill score $(s - b)/(1 - b)$ relative to the best trivial baseline b. The baselines are all-singletons, one big HOG per family, and random relabelling. We report no composite score.

Validation of the metrics. The test suite has 95 tests with hand-derived expected values. Ten deliberate code mutations (for example, swapping speciation and duplication handling) all made the suite fail. On every family, the reference-derived prediction scored 1.0 on all endpoints. When we ran trivial baselines and error analyses on real data, we found five metric defects that toy data had not shown (S2 Text). For example, all-singletons first "recovered" 86% of YGOB WGD events, because the hybrid-origin interval has no upper bound within the family's range and many copies are single genes. We therefore score only events that both baselines can fail. Of 1,592 YGOB WGD events, 202 are testable.

Tool comparison

We ran OrthoFinder 3.1.5 (default and DendroBLAST modes) and FastOMA 0.5.0. Both tools received the same proteomes and species trees. Fig 2 shows the primary endpoints per panel with the best trivial baseline. Per-panel results are descriptive because each panel has two to five headline items. In the pooled confirmatory test (paired permutation, OrthoFinder default versus FastOMA, Holm correction over the four endpoints [43]; Table 3), OrthoFinder scored higher on per-level F1. The other endpoints differed in the same direction, but not significantly. Most of the difference came from panels E and V. OrthoFinder in DendroBLAST mode had the highest

pooled merge-level accuracy (0.886 versus 0.846 for default OrthoFinder and 0.712 for FastOMA; $n = 13$) and event recovery (0.617, 0.479 and 0.194; $n = 11$), but not on every panel.

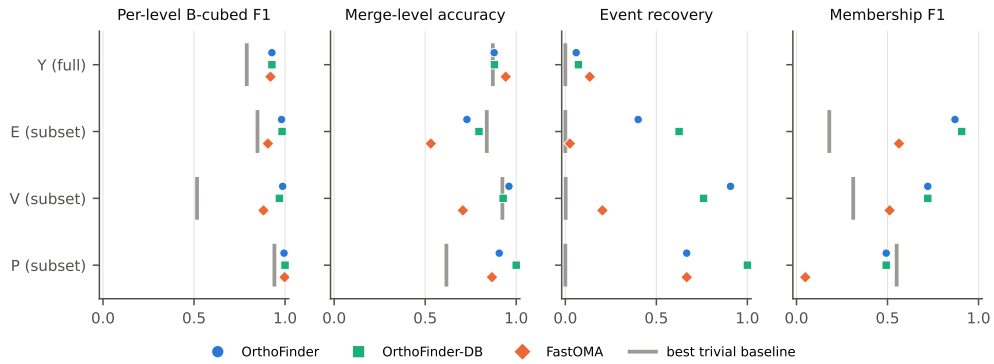


Fig 2. Primary endpoints per panel. Mean over headline items (a collection counts as one item); grey bars show the best trivial baseline. OrthoFinder-DB = OrthoFinder DendroBLAST mode. Number of items per panel: Y 2, E 5 (membership 3), V 3 (events 2, membership 1), P 3 (events 2, membership 1). Panel K has no headline item with asserted pairs. Membership F1 is undefined for Y.

Table 3. Confirmatory comparison, OrthoFinder minus FastOMA. Paired permutation test over pooled headline items (10,000 sign flips), Holm-adjusted over the four endpoints. Panels Y and K in full mode, E, V and P in subset mode.

Endpoint	Items	Mean difference	Holm p
Per-level B-cubed F1	13	+0.054	0.035
Merge-level accuracy	13	+0.134	0.058
Event recovery	11	+0.285	0.12
Membership F1	5	+0.315	0.12

Trivial baselines put the numbers in context. On panel E, one big HOG per family had higher merge-level accuracy (0.838) than both tools (OrthoFinder 0.729, FastOMA 0.531). Most asserted nodes there are speciation nodes, and both tools split speciation pairs (FastOMA 66.7% and OrthoFinder 20.4% of such pairs). FastOMA also merged 24.4% of pairs under duplication nodes too early, mostly at the eukaryote level (OrthoFinder 0.1%). On plant bHLH proteins, FastOMA had high membership precision (0.959) but recall of 0.025: it placed the 435 members in 186 groups.

Phenomena that separate the methods

Hidden paralogy after WGD. In 49 YGOB units, *S. cerevisiae* and *C. glabrata* each kept a single copy, but from different WGD tracks. These genes look like orthologs but are paralogs [39]. OrthoFinder classified 4% of these pairs correctly and FastOMA 12%, compared with 91% and 87% for 1,189 true ortholog pairs (Fig 3A).

WGD events. Over the 201 testable YGOB WGD events, the two tools failed in opposite ways (Fig 3B). OrthoFinder kept ohnolog copies apart in only 26% of events, but on average 79% of each copy's orthologs shared a HOG (minimum over the two copies). FastOMA kept copies apart in 64% of events, with an average of 57%. On vertebrate WGDs (Hox, below) FastOMA was lower on both measures. Within YGOB units, accuracy at WGD nodes was lower than at speciation nodes by 0.68 for

OrthoFinder and 0.21 for FastOMA ($n = 532$ units). We did not test the causes of these errors.

Vertebrate duplications. FastOMA recovered 8% of the 13 Hox 2R events and 9% of the 11 3R events; OrthoFinder recovered 77% and 82% (Fig 3C). FastOMA placed *X. laevis* L/S homeologs correctly in 43% of 333 units and split 56% of them; OrthoFinder placed 99% correctly.



Fig 3. Phenomena. (A) Fraction of yeast gene pairs classified correctly, for true orthologs (same YGOB track) and pseudo-orthologs (single surviving copies on different tracks); 95% bootstrap intervals. (B) Testable YGOB WGD events: fraction with copies in different HOGs at the event level, minimum fraction of each copy's orthologs that share a HOG, and events recovered. (C) Recovery of Hox cluster duplications at the 2R (Gnathostomata) and 3R (Teleostei) levels.

Robustness to data errors

We perturbed the yeast panel and re-ran both tools (Fig 4). Splitting 10% of family members into two fragments lowered merge-level accuracy by 0.050 (OrthoFinder) and 0.114 (FastOMA) when both fragments had to be placed correctly. When one correctly placed fragment was enough, the drops were 0.003 and 0.024. Inserting 20 or 200 human proteins into the *K. lactis* proteome, swapping one contested species-tree node, or removing two pre-WGD species changed the primary endpoints by at most 0.005. In the 200-protein test, OrthoFinder placed 103 and FastOMA 133 of the contaminants into HOGs with yeast reference genes at some level, but the relations among the yeast genes did not change. A repeated OrthoFinder run was identical to the first; a repeated FastOMA run differed in 2% of YGOB units (mean change in merge-level accuracy −0.001, SD 0.038). FastOMA's changes after these perturbations ($\leq$0.004) are of the same order as the shift between two unperturbed FastOMA runs; with one replicate we cannot separate them. The perturbations were mild. Neither tool changed much, but we cannot say how larger errors would affect them.

Scoring on proteome subsets

Full runs on panels E, V and P would have taken several hours per tool, so we scored them on subsets: all proteins with DIAMOND hits [44] to frozen reference members, a limited number of proteins that share only a domain with reference members, and a random background. We tested this choice on panels Y and K, where we ran both modes. For all tools and testable primary endpoints except event recovery on K (a single event), the subset-minus-full difference was within the pre-registered margin of ±0.05 (TOST, 90% paired interval [45]), and 96–99.6% of gene pairs had the same predicted merge level. The test was informative mainly on K (subset 6% of the

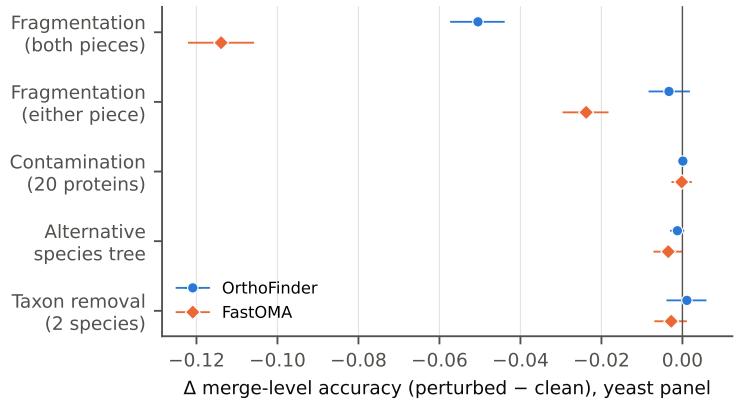


Fig 4. Effect of perturbations on the yeast panel. Change in merge-level accuracy (perturbed minus clean; 95% bootstrap interval over 1,977 units). Fragmentation is scored with two rules: the parent gene counts as correct if both fragments, or either fragment, are placed correctly. The 200-protein contamination test (not shown) gave changes of +0.0004 (OrthoFinder) and −0.0010 (FastOMA).

proteome); on Y the subset covered 54% of the proteome. Results for E, V and P are therefore conditional on subset mode.

Discussion

HOBench shows that references built from published sources can test hierarchical orthology at specific levels and for specific events. The hidden-paralogy result is the clearest example: both tools score well on true yeast orthologs and poorly on pseudo-orthologs. A score that pooled these pairs would be dominated by the 1,189 true ortholog pairs and would hide the failure.

The benchmark has clear limits. First, the number of independent families is small: 13 headline items, with two to five per panel. Only the pooled test is inferential, and its power is low. Second, references are incomplete by design. Many families are partly unresolved, and membership F1 is defined for only five items. Third, the families are biased toward well-studied groups with machine-readable data. No headline family covers incomplete lineage sorting, gene fusion or a fast-evolving lineage, and only one headline family (Y03) scores a transfer; P04 tags its HGT but leaves it unresolved. Fourth, some choices are ours and debatable. The interval for the yeast WGD and copies that consist of a single gene make 87% of YGOB events untestable; the Hox reference relies on curated nomenclature, and two prokaryote families locate proteins through COG membership [34]. Fifth, FastOMA assigns proteins with OMAmer [46], whose database is built from OMA HOGs that may include our species; we could not remove this possible advantage within the budget. Sixth, we compared two tools in two OrthoFinder modes; agreement analyses with more methods would be more informative. Seventh, panels E, V and P were scored in subset mode, which we validated only on Y and K.

What the agent experiment showed. The agents completed the survey, curation, software and analysis in about 21 hours of wall-clock time, of which about 11 hours were lost when we hit the account's usage limit three times. The most useful process elements were independent design reviews, which found real defects before scoring, mechanical checks of provenance, and trivial baselines and error analyses on real data,

which exposed five metric defects. The main failures were launching too many subagents at once (which triggered the first outage), late checks of data sources (a tool-derived table was found only during curation), and accepting metric definitions before running controls. An independent agent fact-checked the first report against the result files and found eight wrong numbers and several overstatements, which we corrected. Because no human reviewed the curated references line by line, errors may remain; the blind re-derivation bounds the error rate per family only roughly (at most 1 of 30 items wrong for passing families). S1 Text lists the main ideas that did not work.

Methods

Agent setup

The project ran in Claude Code, an agent harness, on one compute node (12 cores, about 50 GB usable memory, no job scheduler) with internet access. A PI agent received the project prompt and a goal message (both in the repository, `prompts/`) and ran autonomously for about 21 hours within a 24-hour budget. It used Anthropic Claude models: a larger model for the PI, the literature survey, core software, curation, design review and analysis, and a smaller, faster model for routine engineering, verification, audits and fact-checking. The PI agent wrote the research record (`RESEARCH_TREE.md`, decisions D0–D12), process rules (`LESSONS.md`) and the design document, and it launched 31 subagents: 4 literature surveyors, 4 software engineers (metrics, curation tools, provenance checks, tool wrapper), 1 panel and data engineer, 10 curators (two to six families each), 1 evaluator, 1 analyst, 1 blind verifier, 1 mutation tester, 1 catalogue builder, 1 release engineer, 1 design-review coordinator (which launched 7 reviewers), 2 second-round reviewers, 2 process reviewers and 1 fact-checker. Subagents received written briefs and shared files through the project directory and git. They returned short summaries, and the PI read these summaries rather than raw logs. Subagents could not see each other's work unless it was committed. Reviewers and verifiers did not see the work they checked until they had written their own assessments.

Humans intervened in three ways: they wrote the prompts, they supplied 49 of 50 paywalled papers or supplements that the agents listed in a ranked request file, and after outages they typed "Try again" or "Please continue after stop". This manuscript and the repository were written by the same PI agent after a second prompt (in the repository).

Literature and data survey

Four subagents surveyed (i) orthology benchmarks and metrics, (ii) candidate families in animals and fungi, (iii) candidate families in plants, protists and prokaryotes, and (iv) proteome sources, species panels and species trees. They verified every citation against Europe PMC or Crossref and opened supplements to confirm that member lists existed. The surveys produced 85 candidate families with phenomenon tags and data readiness, and a ranked list of 50 papers that could not be retrieved automatically.

Design and review

The PI wrote a design document, which seven independent reviewer agents (external PI, prior work, statistics, data, algorithms and compute, AI-agent failure modes, molecular evolution) reviewed. A coordinator merged their proposals. The PI

accepted the synthesis, which changed the merge levels in the reference from points to intervals, introduced two-sided event recovery and membership F1, replaced an incorrect decoy set (β-lactamase classes A, C and D are homologous) with non-homologous isofunctional enzymes [38], and added mechanical provenance checks. A second round (two reviewers) reviewed the design together with first results. Table 4 lists the main decisions.

Table 4. Main design decisions.

Decision	Reason	Consequence
Score only asserted relations; unresolved = U	absence of a statement is not evidence	partial references are usable; fewer scored pairs
Merge levels in the reference as intervals	contested timing, allopolyploidy, HGT	correct WGD predictions were misscored with point values
Tool-derived sources excluded	avoid circularity	Braasch 2016 3R ohnologs dropped (Ensembl Compara)
Members extracted by parsers; quotes checked by script	prevent invented identifiers	slower curation
Same-species mapping, ≥98% identity; ≥80% mapped for headline	wrong paralog or homeolog otherwise	6 families secondary
Allopolyploid interval for the yeast WGD	hybrid origin [19]	87% of YGOB events untestable
Score only testable events (both baselines can fail)	trivial predictors scored 0.86	202 of 1,592 YGOB events scored
Leaf-level duplications reported separately	one big HOG recovers them correctly	headline events = internal branches
Five panels; full mode for Y, K; subsets for E, V, P	compute budget	subset validated on Y, K only
Freeze references before scoring large panels	avoid tuning to results	corrections logged as errata
One pooled confirmatory test; panels descriptive	2–5 items per panel	low power

Species panels

Proteomes are UniProt reference proteomes, release 2026_03 [16], taken as one canonical protein per gene from the UniProt FTP site and pinned by sha256. Panel species were chosen to cover each panel's events and to overlap with species sampled in the source papers. Species trees were taken from published phylogenomic studies, made binary and given level names; contested nodes are documented with citations in `panels/<P>/levels.tsv`, and each panel has one alternative topology.

Curation protocol

Curators followed a written protocol (repository, `docs/CURATION_PROTOCOL.md`). They could not open tool outputs. Each family has a `build.py` that reads stored source files and writes the member table, reference tree, mapping report and assertion table. Assertions have three tiers: T1, stated in the text; T2, derived from curated labels by fixed rules; T3, read from published trees with support ≥70% bootstrap or ≥0.95 posterior. Headline numbers use T1 and T2. We did not apply the rule "same label and at most one gene per species means orthologs" (an S node) across WGD lineages unless the set was synteny-based, because reciprocal loss creates pseudo-orthologs. When a source gave only a domain sequence (kinesin and myosin motor domains), mapping required a domain of ≥250 residues covered to ≥90% at

≥98% identity. A script checked that each quote occurs in the stored source, that each gene is in the pinned proteome, and that thresholds hold. A blind verifier re-derived a random sample of up to 30 assertions per family (fixed seed) from the source; two or more discrepancies failed the family (Clopper–Pearson bounds reported [47]). A script searched all family files for tool-derived sources and references to tool outputs and found none after review of flagged lines.

Tool runs

A Python wrapper ran OrthoFinder 3.1.5 (`-s` with the panel tree; default and `-M dendroblast`) and the FastOMA 0.5.0 Nextflow workflow with the same tree and precomputed OMAmer placements (LUCA database) and recorded commands, versions, input checksums and run times. A flat baseline used OrthoFinder orthogroups without hierarchy. In OrthoXML, levels were matched to panel levels by species content, and duplications were placed at the level declared by the enclosing group (a later-placement reading was a sensitivity analysis, following pyHam [48]). Subsets were built once with DIAMOND (`--more-sensitive`, $E \leq 10^{-5}$) from frozen member lists and stored as identifier lists.

Statistics

The unit of analysis is a family; a collection counts as one item, with its units aggregated first. Per-panel results are descriptive. The confirmatory test is a paired sign-flip permutation test (10,000 permutations, fixed seed) per primary endpoint over pooled headline items, with Holm correction [43]. Agreement between methods uses adjusted mutual information [49]. Robustness intervals are bootstrap intervals over units. The subset-versus-full test is a two one-sided tests procedure with a margin of 0.05 fixed before the runs.

Ideas taken from the literature

The HOG definition and its equivalence to reconciled trees come from OMA work [1, 2]; the method for inducing levels in OrthoXML from pyHam [48]; family-level fission and fusion measures from OrthoBench [8, 9]; the use of expert reconciled trees as references from SwissTree [10]; the benchmark-service framing from QfO [6, 7]; the theory of orthology relations from [40, 41]; hidden paralogy and synteny tracks from [17, 18, 39]; the hybrid WGD model from [19]; analogous enzymes as merge decoys from [38]; B-cubed, adjusted mutual information, TOST, Holm and Clopper–Pearson methods from [42, 43, 45, 47, 49]. Each reference family uses the classifications and phylogenies of the papers cited in Table 2 and in its `family.yaml`.

Data and code availability

The repository contains the `hobench` package, scripts to score a new method (`score_method.py`), frozen references with provenance, panel definitions, subset identifier lists, the outputs of all runs reported here, result tables, the prompts and the complete process record. The proteomes used are provided as a release archive (UniProt, CC BY 4.0). [Repository URL and archive DOI to be added by the maintainer.]

Supporting information

S1 Text. Ideas that did not work. Major approaches that the agents tried and abandoned or corrected, with reasons.

S2 Text. Metric errata and verification. The five metric defects found by controls and error analysis, the post-freeze reference corrections, and blind-verification results per family (repository files `results/ERRATA.md`, `families/ERRATA.md`, `families/VERIFICATION.md`).

S1 Table. Catalogue of all 32 reference families and collections with sources, mapping rates, events by tier, verification status and abandoned candidates (repository `results/catalogue/`).

Acknowledgments

We thank the authors of the cited studies, whose published classifications and phylogenies form the references.

References

1. Altenhoff AM, Gil M, Gonnet GH, Dessimoz C. Inferring hierarchical orthologous groups from orthologous gene pairs. PLoS ONE. 2013;8:e53786. doi:10.1371/journal.pone.0053786.
2. Train CM, Glover NM, Gonnet GH, Altenhoff AM, Dessimoz C. Orthologous Matrix (OMA) algorithm 2.0: more robust to asymmetric evolutionary rates and more scalable hierarchical orthologous group inference. Bioinformatics. 2017;33:i75-82. doi:10.1093/bioinformatics/btx229.
3. Emms DM, Kelly S. OrthoFinder: phylogenetic orthology inference for comparative genomics. Genome Biology. 2019;20:238. doi:10.1186/s13059-019-1832-y.
4. Emms DM, et al. OrthoFinder: improved phylogenetic orthology inference with enhanced accuracy and scalability. Nature Methods. 2026. doi:10.1038/s41592-026-03126-6.
5. Majidian S, Nevers Y, et al. Orthology inference at scale with FastOMA. Nature Methods. 2025. doi:10.1038/s41592-024-02552-8.
6. Altenhoff AM, Boeckmann B, Capella-Gutierrez S, et al. Standardized benchmarking in the quest for orthologs. Nature Methods. 2016;13:425-30. doi:10.1038/nmeth.3830.
7. Nevers Y, Jones TEM, Jyothi D, et al. The Quest for Orthologs orthology benchmark service in 2022. Nucleic Acids Research. 2022;50:W623-32. doi:10.1093/nar/gkac330.
8. Trachana K, Larsson TA, Powell S, et al. Orthology prediction methods: a quality assessment using curated protein families. BioEssays. 2011;33:769-80. doi:10.1002/bies.201100062.

9. Emms DM, Kelly S. Benchmarking orthogroup inference accuracy: revisiting Orthobench. Genome Biology and Evolution. 2020;12:2258-66. doi:10.1093/gbe/evaa211.

10. Boeckmann B, Robinson-Rechavi M, Xenarios I, Dessimoz C. Conceptual framework and pilot study to benchmark phylogenomic databases based on reference gene trees. Briefings in Bioinformatics. 2011;12:423-35. doi:10.1093/bib/bbr034.

11. Dalquen DA, Anisimova M, Gonnet GH, Dessimoz C. ALF—a simulation framework for genome evolution. Molecular Biology and Evolution. 2012;29:1115-23. doi:10.1093/molbev/msr268.

12. Sarton-Lohéac G, et al. Reconstructing evolutionary histories with hierarchical orthologous groups. Journal of Molecular Evolution. 2025. doi:10.1007/s00239-025-10277-1.

13. Shen XX, Opulente DA, Kominek J, et al. Tempo and mode of genome evolution in the budding yeast subphylum. Cell. 2018;175:1533-45. doi:10.1016/j.cell.2018.10.023.

14. Coleman GA, Davín AA, Mahendrarajah TA, et al. A rooted phylogeny resolves early bacterial evolution. Science. 2021;372:eabe0511. doi:10.1126/science.abe0511.

15. Burki F, Roger AJ, Brown MW, Simpson AGB. The new tree of eukaryotes. Trends in Ecology & Evolution. 2020;35:43-55. doi:10.1016/j.tree.2019.08.008.

16. The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2025. Nucleic Acids Research. 2025;53:D609-17. doi:10.1093/nar/gkae1010.

17. Byrne KP, Wolfe KH. The Yeast Gene Order Browser: combining curated homology and syntenic context reveals gene fate in polyploid species. Genome Research. 2005;15:1456-61. doi:10.1101/gr.3672305.

18. Scannell DR, Frank AC, Conant GC, Byrne KP, Woolfit M, Wolfe KH. Independent sorting-out of thousands of duplicated gene pairs in two yeast species descended from a whole-genome duplication. Proceedings of the National Academy of Sciences USA. 2007;104:8397-402. doi:10.1073/pnas.0608218104.

19. Marcet-Houben M, Gabaldón T. Beyond the whole-genome duplication: phylogenetic evidence for an ancient interspecies hybridization in the baker's yeast lineage. PLoS Biology. 2015;13:e1002220. doi:10.1371/journal.pbio.1002220.

20. Hall C, Brachat S, Dietrich FS. Contribution of horizontal gene transfer to the evolution of *Saccharomyces cerevisiae*. Eukaryotic Cell. 2005;4:1102-15. doi:10.1128/EC.4.6.1102-1115.2005.

21. Gojković Z, Knecht W, Zameitat E, et al. Horizontal gene transfer promoted evolution of the ability to propagate under anaerobic conditions in yeasts. Molecular Genetics and Genomics. 2004;271:387-93. doi:10.1007/s00438-004-0995-7.

22. Wickstead B, Gull K, Richards TA. Patterns of kinesin evolution reveal a complex ancestral eukaryote with a multifunctional cytoskeleton. BMC Evolutionary Biology. 2010;10:110. doi:10.1186/1471-2148-10-110.

23. Vargová R, Wideman JG, Derelle R, et al. A eukaryote-wide perspective on the diversity and evolution of the ARF GTPase protein family. Genome Biology and Evolution. 2021;13:evab157. doi:10.1093/gbe/evab157.

24. Klöpper TH, Kienle N, Fasshauer D, Munro S. Untangling the evolution of Rab G proteins: implications of a comprehensive genomic analysis. BMC Biology. 2012;10:71. doi:10.1186/1741-7007-10-71.

25. Findeisen P, Mühlhausen S, Dempewolf S, Hertzog J, Zietlow A, Carlomagno T, et al. Six subgroups and extensive recent duplications characterize the evolution of the eukaryotic tubulin protein family. Genome Biology and Evolution. 2014;6:2274-88. doi:10.1093/gbe/evu187.

26. Hirst J, Schlacht A, Norcott JP, et al. Characterization of TSET, an ancient and widespread membrane trafficking complex. eLife. 2014;3:e02866. doi:10.7554/eLife.02866.

27. Lagman D, Ocampo Daza D, Widmark J, Abalo XM, Sundström G, Larhammar D. The vertebrate ancestral repertoire of visual opsins, transducin alpha subunits and oxytocin/vasopressin receptors was established by duplication of their shared genomic region in the two rounds of early vertebrate genome duplications. BMC Evolutionary Biology. 2013;13:238. doi:10.1186/1471-2148-13-238.

28. Holland PWH, Booth HAF, Bruford EA. Classification and nomenclature of all human homeobox genes. BMC Biology. 2007;5:47. doi:10.1186/1741-7007-5-47.

29. Session AM, Uno Y, Kwon T, et al. Genome evolution in the allotetraploid frog *Xenopus laevis*. Nature. 2016;538:336-43. doi:10.1038/nature19840.

30. Pires N, Dolan L. Origin and diversification of basic-helix-loop-helix proteins in plants. Molecular Biology and Evolution. 2010;27:862-74. doi:10.1093/molbev/msp288.

31. Yoshida H, Tanimoto E, Hirai T, et al. Evolution and diversification of the plant gibberellin receptor GID1. Proceedings of the National Academy of Sciences USA. 2018;115:E7844-53. doi:10.1073/pnas.1806040115.

32. Danielson J H, Johanson U. Unexpected complexity of the aquaporin gene family in the moss *Physcomitrella patens*. BMC Plant Biology. 2008;8:45. doi:10.1186/1471-2229-8-45.

33. Yutin N, Puigbò P, Koonin EV, Wolf YI. Phylogenomics of prokaryotic ribosomal proteins. PLoS ONE. 2012;7:e36972. doi:10.1371/journal.pone.0036972.

34. Galperin MY, Wolf YI, Makarova KS, Vera Alvarez R, Landsman D, Koonin EV. COG database update: focus on microbial diversity, model organisms, and widespread pathogens. Nucleic Acids Research. 2021;49:D274-81. doi:10.1093/nar/gkaa1018.

35. Baldauf SL, Palmer JD, Doolittle WF. The root of the universal tree and the origin of eukaryotes based on elongation factor phylogeny. Proceedings of the National Academy of Sciences USA. 1996;93:7749-54. doi:10.1073/pnas.93.15.7749.

36. Iwabe N, Kuma Ki, Hasegawa M, Osawa S, Miyata T. Evolutionary relationship of archaebacteria, eubacteria, and eukaryotes inferred from phylogenetic trees of duplicated genes. Proceedings of the National Academy of Sciences USA. 1989;86:9355-9. doi:10.1073/pnas.86.23.9355.

37. Liu Y, Richards TA, Aves SJ. Ancient diversification of eukaryotic MCM DNA replication proteins. BMC Evolutionary Biology. 2009;9:60. doi:10.1186/1471-2148-9-60.

38. Omelchenko MV, Galperin MY, Wolf YI, Koonin EV. Non-homologous isofunctional enzymes: a systematic analysis of alternative solutions in enzyme evolution. Biology Direct. 2010;5:31. doi:10.1186/1745-6150-5-31.

39. Scannell DR, Byrne KP, Gordon JL, Wong S, Wolfe KH. Multiple rounds of speciation associated with reciprocal gene loss in polyploid yeasts. Nature. 2006;440:341-5. doi:10.1038/nature04562.

40. Hellmuth M, Hernandez-Rosales M, Huber KT, Moulton V, Stadler PF, Wieseke N. Orthology relations, symbolic ultrametrics, and cographs. Journal of Mathematical Biology. 2013;66:399-420. doi:10.1007/s00285-012-0525-x.

41. Lafond M, El-Mabrouk N. Orthology and paralogy constraints: satisfiability and consistency. BMC Genomics. 2014;15:S12. doi:10.1186/1471-2164-15-S6-S12.

42. Amigó E, Gonzalo J, Artiles J, Verdejo F. A comparison of extrinsic clustering evaluation metrics based on formal constraints. Information Retrieval. 2009;12:461-86. doi:10.1007/s10791-008-9066-8.

43. Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics. 1979;6:65-70.

44. Buchfink B, Reuter K, Drost HG. Sensitive protein alignments at tree-of-life scale using DIAMOND. Nature Methods. 2021;18:366-8. doi:10.1038/s41592-021-01101-x.

45. Schuirmann DJ. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. Journal of Pharmacokinetics and Biopharmaceutics. 1987;15:657-80. doi:10.1007/BF01068419.

46. Rossier V, Warwick Vesztrocy A, Robinson-Rechavi M, Dessimoz C. OMAmer: tree-driven and alignment-free protein assignment to subfamilies outperforms closest sequence approaches. Bioinformatics. 2021;37:2866-73. doi:10.1093/bioinformatics/btab219.

47. Clopper CJ, Pearson ES. The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika. 1934;26:404-13. doi:10.1093/biomet/26.4.404.

48. Train CM, Pignatelli M, Altenhoff A, Dessimoz C. iHam and pyHam: visualizing and processing hierarchical orthologous groups. Bioinformatics. 2019;35:2504-6. doi:10.1093/bioinformatics/bty994.

49. Vinh NX, Epps J, Bailey J. Information theoretic measures for clusterings comparison: variants, properties, normalization and correction for chance. Journal of Machine Learning Research. 2010;11:2837-54.